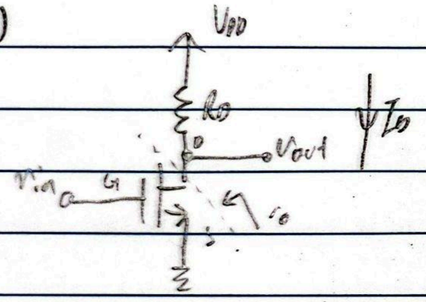
Chai Zi Yang A0262349Y Tutorial 1

Question 1

1)



i) $A_v = g_m r_o$ ← defined by question (even though should have -ve sign)

$= 2\sqrt{\frac{1}{2}\left(\frac{W}{L}\right)k_n' I_D} \times \frac{1}{\lambda I_D}$

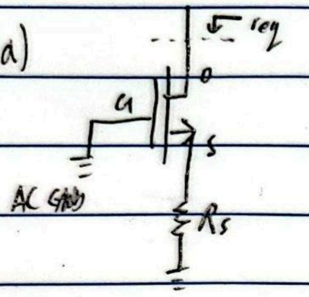
$= \sqrt{2\left(\frac{W}{L}\right)k_n' \frac{1}{I_D}} \times \frac{1}{\lambda}$ #

ii) increasing I_D, increases g_m in $\sim\sqrt{I_D}$, but decreases r_o in $\frac{1}{I_D}$

∴ A_v cannot be increased by increasing I_D

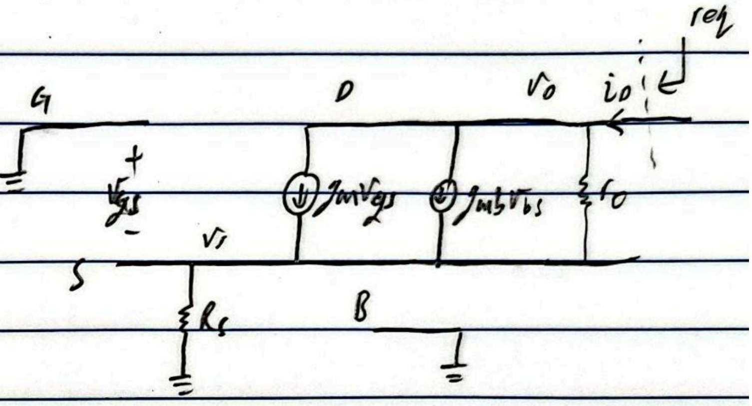
Question 2

2)a)



Assume in saturation

Assume body is grounded



$v_{gs} = v_{bs} = 0 - v_s = -v_s$

KCL at D: $i_o = g_m v_{gs} + g_{mb} v_{bs} + \frac{v_o - v_s}{r_o} = v_s / R_s$ $\qquad r_{eq} = \frac{\partial v_o}{\partial i_o}$

$i_o = -(g_m + g_{mb}) v_s + \frac{v_o - v_s}{r_o} = \frac{v_s}{R_s}$

↓ ∂v_o

$\frac{\partial i_o}{\partial v_o} = -(g_m + g_{mb})\frac{\partial v_s}{\partial v_o} + \frac{1}{r_o} - \frac{1}{r_o}\frac{\partial v_s}{\partial v_o} = \frac{\partial v_s}{\partial v_o}\frac{1}{R_s}$

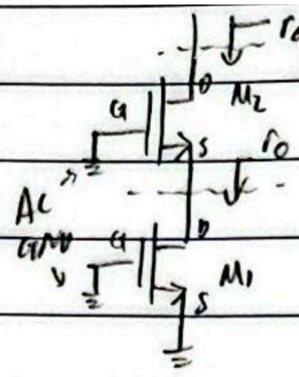
$\frac{\partial v_s}{\partial v_o} = \dfrac{\frac{1}{r_o}}{\frac{1}{R_s} + (g_m + g_{mb}) + \frac{1}{r_o}}$

$= \dfrac{R_s}{R_s + r_o + R_s r_o (g_m + g_{mb})}$

$r_{eq} = \frac{\partial v_o}{\partial i_o} = R_s \times \frac{\partial v_o}{\partial v_s}$

$= R_s + r_o + R_s r_o (g_m + g_{mb})$ #

2b)

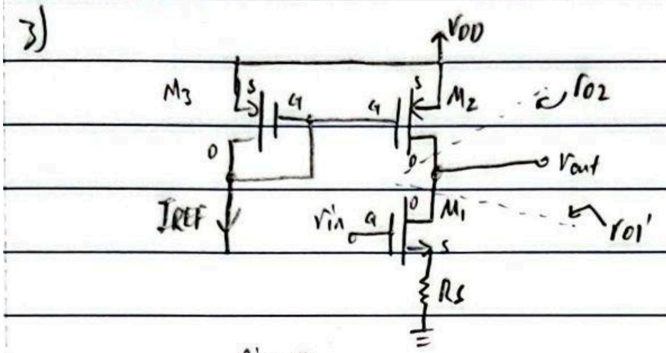


from part a),

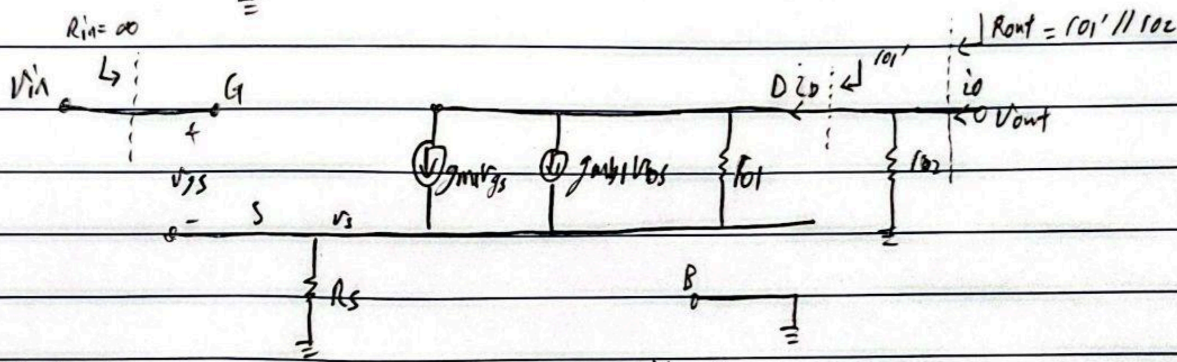
$r_{eq} = r_o + r_{o2} + r_o r_{o2}(g_{m2} + g_{mb2})$ #

Question 3

3)



Assume body is tied to GND for M_1 & V_{DD} for $M_2 M_3$



$$r_{o1}' = \frac{\partial v_{out}}{\partial i_o}\Big|_{v_{in}=0}$$

from 2a) $\Rightarrow r_{o1}' = R_s + r_{o1} + R_s r_{o1}(g_{m1}+g_{mb1})$

$$G_M = \frac{\partial i_o}{\partial v_{in}}\Big|_{v_{out}=0}$$

$$i_o = g_{m1} v_{gs} + g_{mb1} v_{bs} + \frac{v_{out}-v_s}{r_{o1}} + \frac{v_{out}}{r_{o2}}$$

$$i_o = g_{m1}(v_{in}-v_s) + g_{mb1}(0-v_s) + \frac{v_{out}-v_s}{r_{o1}} - \frac{v_{out}}{r_{o2}}$$

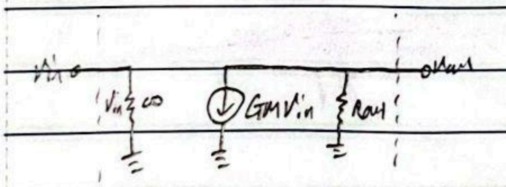
$\downarrow \partial v_{in}$ $\quad v_{out}=0$

since $v_{out} = 0$

$$\frac{\partial i_o}{\partial v_{in}} = g_{m1} - g_{m1}\frac{\partial v_s}{\partial v_{in}} - g_{mb1}\frac{\partial v_s}{\partial v_{in}} - \frac{1}{r_{o1}}\frac{\partial v_s}{\partial v_{in}} + 0 = \frac{\partial v_s}{\partial v_{in}}\frac{1}{R_s}$$

$$\frac{\partial v_s}{\partial v_{in}} = \frac{g_{m1}}{g_{m1}+g_{mb1}+\frac{1}{r_{o1}}+\frac{1}{R_s}}$$

$$\therefore G_M = \frac{r_{o1} g_{m1}}{r_{o1} R_s (g_{m1}+g_{mb1}) + r_{o1} + R_s}$$



$$A_v = \frac{v_{out}}{v_{in}} = -G_M R_{out}$$

$$= \frac{-r_{o1} g_{m1}}{r_{o1}R_s(g_{m1}+g_{mb1}) + r_{o1} + R_s} \times \frac{r_{o1}' r_{o2}}{r_{o1}' + r_{o2}}$$

$$= -r_{o1} g_{m1} \times \frac{r_{o2}}{r_{o1}' + r_{o2}}$$

$$= \frac{-r_{o1} r_{o2} g_{m1}}{R_s + r_{o1} + r_{o2} + R_s r_{o1}(g_{m1}+g_{mb1})} \quad \#$$